

Taxing Income when Income Risk is Endogenous

Paweł Doligalski

University of Bristol

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- Government set a (nonlinear) income tax $T(z)$.
- Agent θ chooses labour ℓ , earns $z = \theta\ell$ and consumes $c = z - T(z)$.
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What are the incentives to manipulate risk?

What are the implications for the design of tax-and-transfer systems?

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and Skov (2014a,b)...
4. More generally, many decisions affect income risk (but typically also
return): education, occupation, financial investments

Plan of the presentation

1. Theory: incentives to manipulate income risk & implications for tax design
 - Risk manipulation limits how quickly MTRs can decline with income
2. Quantification: a model calibrated to the U.S. income distribution
 - Risk manipulation leads to lower transfers that are phased-out more slowly

Related literature

- **Moral hazard with risk manipulation:** Diamond (1998); Barron, Georgiadis, and Swinkels (2020); Kocherlakota (2025)
- **Income randomisation as tax avoidance:** Landier and Plantin (2017)
- **Robustness with respect to production technologies:** Carroll (2015); Vairo (2026)
- **Income taxes in settings with endogenous income risk**
 - **portfolio choice:** Domar and Musgrave (1944); Feldstein (1969); Kanbur (1979); Stiglitz (1982)
 - **performance pay:** Doligalski, Ndiaye, and Werquin (2023); Ferey, Haufler, and Perroni (2023); Ábrahám, Doligalski, Forstner, and Grübener (2026)
 - **occupational choice:** Cubas and Silos (2017, 2020)
 - ...

Framework

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- distribution of earnings s.t. mean earnings being equal $\theta\ell$.

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A tax-and-transfer schedule based on realised earnings z is $T(z)$.

The implied consumption function (after-tax income) is $C(z) := z - T(z)$.

The agent solves

$$U(\theta) = \sup_{\ell, \mu} \int_0^\infty u(C(z)) d\mu(z) - g(\ell) \quad \text{s.t.} \quad \int_0^\infty z d\mu(z) = \theta \ell \quad (P_\theta)$$

where

- μ is a probability measure (on the Borel space $(\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+))$),
- $C(z)$ is the consumption at earnings z ,
- the constraint ensures that mean earnings are equal $\theta \ell$.

Incentives to manipulate risk

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Proposition 1

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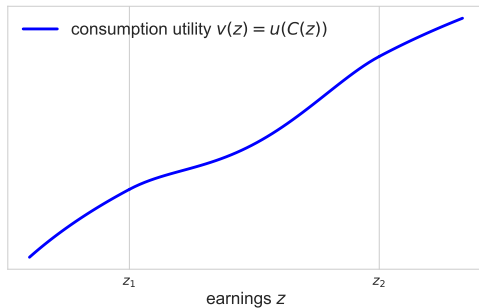
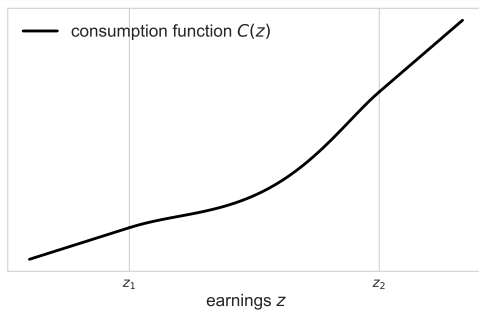
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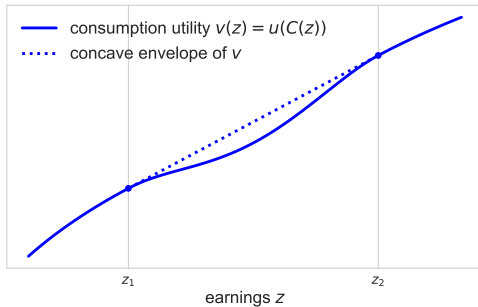
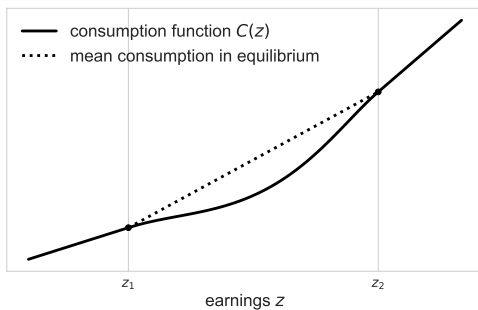
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Take-away: there are incentives for extra risk when $C(z)$ is "convex enough" to make $u(C(z))$ non-concave.

Example: $u(c) = \log c$



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Under differentiability, concavity of $v(z)$ translates into

$$-\frac{u''(C(z))}{u'(C(z))} \geq \frac{C''(z)}{(C'(z))^2} = -\frac{T''(z)}{(1 - T'(z))^2}, \quad \forall z.$$

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Empirically, marginal tax rates often decline (discontinuously, $T'' = -\infty$) at incomes where the welfare benefits and tax credits are fully phased-out. 

Implications for the tax design

Preliminaries:

- Let $\theta \sim F$.
- Fix a tax function T and let $\{\ell_\theta, \mu_\theta\}$ solve the agent's problem P_θ given T , for all θ . The (aggregate) tax revenue is $\int \int T(z) d\mu(\theta)(z) dF(\theta)$.
- Assume regularity (tax revenue is finite).

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Intuition: A tax schedule that induces agents to take extra risk is Pareto inefficient—agents increase consumption risk only to pay lower taxes.

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Finally, the aggregate consumption falls (risk-averse agents need less resources to obtain the same utility without risk). Thus, the tax revenue increases.

Let the planner have a weighted utilitarian objective, with weights $\omega(\theta)$.

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At any income level z where $v''(z) < 0$, the optimal tax schedule follows the Diamond-Saez formula:

$$\frac{T'(z)}{1 - T'(z)} = \frac{1}{\varepsilon_{z,1-\tau}(z)} \frac{1}{zh(z)} \int_z^\infty (1 - m(y)) e^{\int_z^y (1 - T'(x)) \gamma(x) dx} dH(y).$$

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- Assumed T continuously diff. and piecewise twice-diff.—rules out bunching.
- Notation: the cdf and density of earnings $H(z)$, $h(z)$; the elasticity of earnings $\varepsilon_{z,1-\tau}(z)$; the marginal social welfare weight $m(z(\theta)) \propto \omega(\theta) u'(c(\theta))$.

Extensions:

- Multiple time periods and stochastic shocks
- Alternative dynamic interpretation: intertemporal shifting of income
- Extra frictions (e.g., moral hazard)
- Costly risk manipulation

Assumptions on preferences:

- Common labour disutility: can be relaxed to (multidimensional) heterogeneity.
- Common risk preferences: important and cannot be easily relaxed.
- Additive separability: can allow $u(c)d(\ell)$, but unlikely beyond that.

Quantitative exercise

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Broad idea: fix preferences and find productivity distribution that replicates the observed labour income distribution.

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- Income distributed as Double Pareto-Lognormal:

$$\log(z) = \mu + \sigma N + E_1/\alpha - E_2/\beta,$$

- $N \sim N(0, 1)$, $E_i \sim \text{Exp}(1)$, independent

- MLE tails coefficients: $\alpha = 2.1$, $\beta = 1.4$

► tail figure

► Robustness check (CPS)

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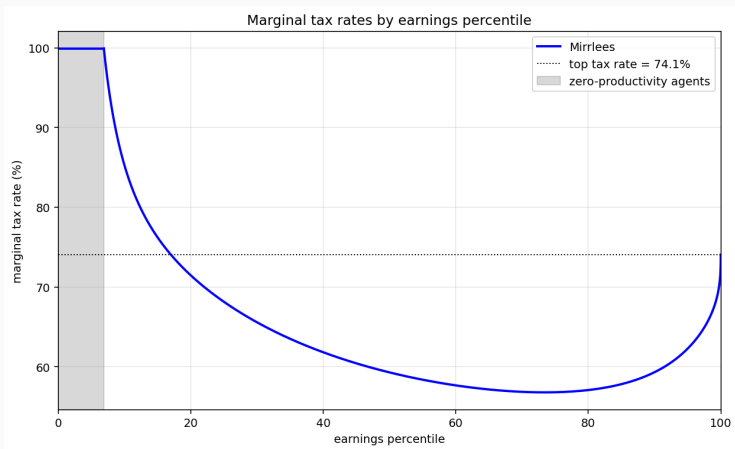
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- MLE tails coefficients: $\alpha = 2.1, \beta = 1.4$ [▶ tail figure](#) [▶ Robustness check \(CPS\)](#)
- 6.9% of households with no labour income $\rightarrow$ assign 0 productivity.

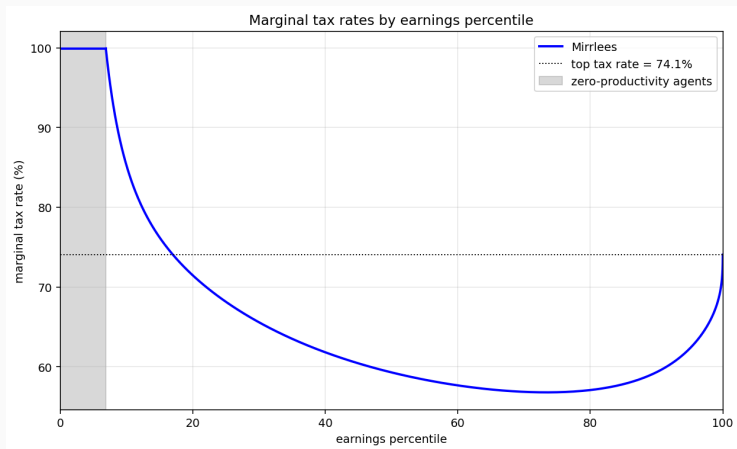
I compare the optimal tax-and-transfer schedule under two scenarios:

1. "Mirrlees": risk manipulation is exogenously ruled out;
2. "endogenous risk": risk manipulation is possible, but optimally prevented.

Utilitarian planner, $u(c) = \log c$

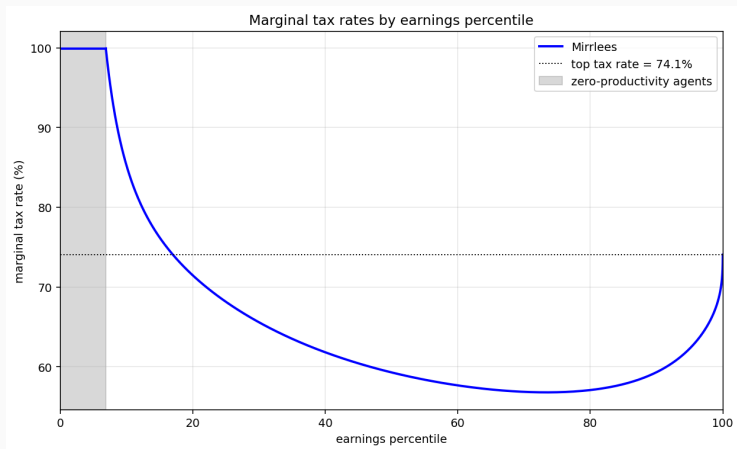


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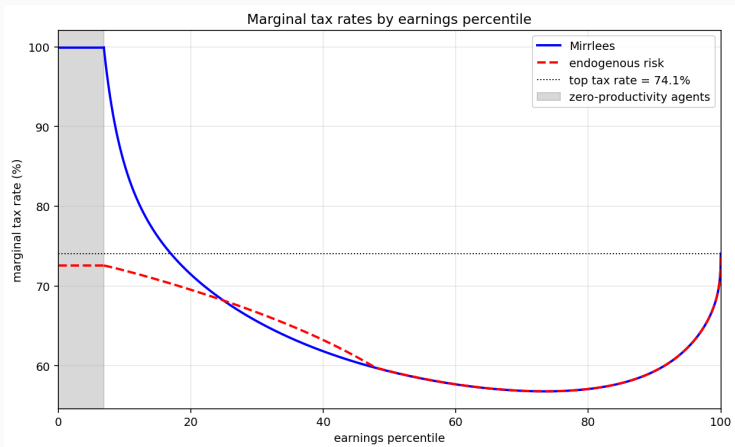
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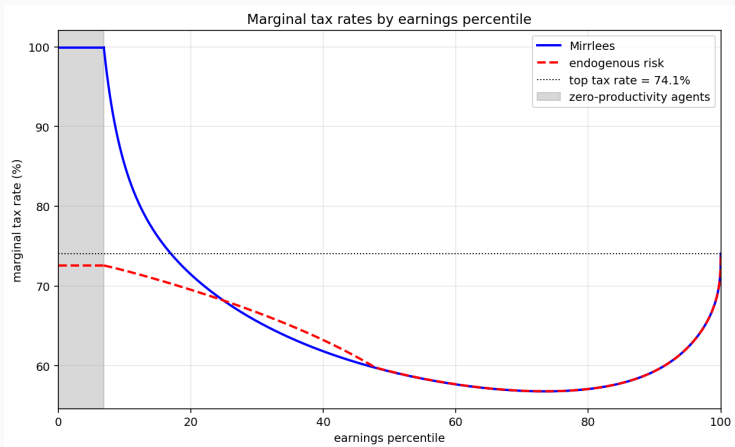


- Very high and strongly declining MTRs at the bottom—high lump-sum transfer that is quickly phased out.
- Importance of thick lower tail and the mass of $\theta = 0$ agents
 - Heathcote and Tsujiyama (2021): thin lower tail and no mass at $\theta = 0 \rightarrow$ low MTRs at the bottom.

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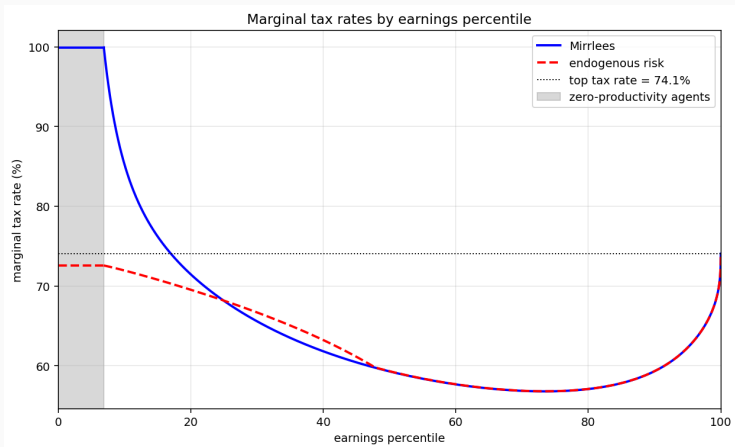


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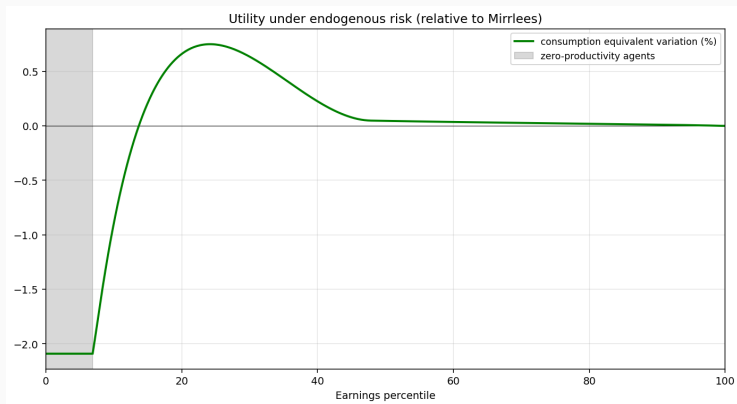
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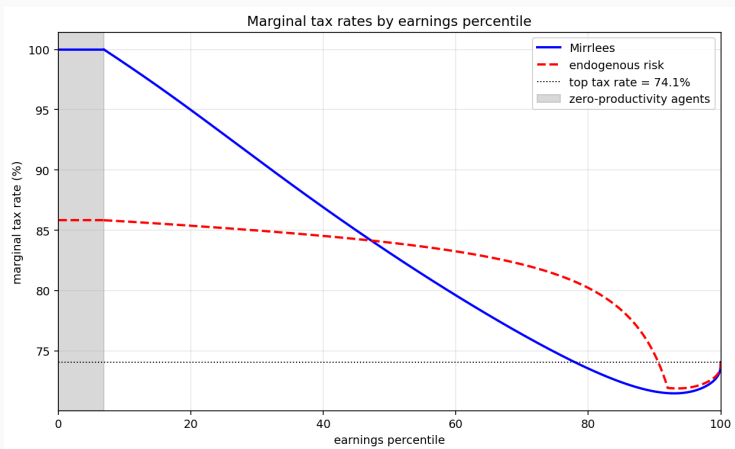
- With risk manipulation, MTRs cannot decline that quickly—lower transfer that is phased out at the lower rate
- MTR at the bottom falls by 27 pp; no change above the median earnings

Utilitarian planner, $u(c) = \log c$,



- Bottom agents lose due to lower transfer, agents in the 15-40 percentile gain due to a slower phase-out of the transfer.
- Overall welfare loss is tiny: -0.046% CEV.

Rawlsian planner, $u(c) = \log c$,



- Mirrlees: MTRs decrease with income over most of the distribution.
- With risk manipulation, the schedule of MTRs substantially flattened.
- Overall welfare loss is larger: transfer to $\theta = 0$ agents falls by -2.24% .

Table 1: Difference in marginal tax rate at the bottom (relative to Mirrlees, pp)

	$\sigma = 0$	$\sigma = 0.5$	$\sigma = 1$	$\sigma = 2$
utilitarian	0	-45	-27	-1
utilitarian (weighted ¹)	-62	-33	-21	-1
Rawlsian	-33	-22	-14	-1

- Largest impact when risk aversion is low.
- Under risk neutrality, optimal tax schedules that prevent risk manipulation are fully linear.

¹Welfare weights follow $\omega(\theta) \propto F(\theta)^{-\frac{1}{2}}$.

Table 2: Welfare loss due to risk manipulation (relative to Mirrlees, CEV)

	$\sigma = 0$	$\sigma = 0.5$	$\sigma = 1$	$\sigma = 2$
utilitarian	0%	-0.05%	-0.05%	-0.002%
utilitarian (weighted ²)	-1.3%	-0.5%	-0.3%	-0.1%
Rawlsian	-9.9%	-4.1%	-2.2%	-0.9%

- Welfare loss is increasing in the strength of redistributive preferences and decreasing in risk aversion.

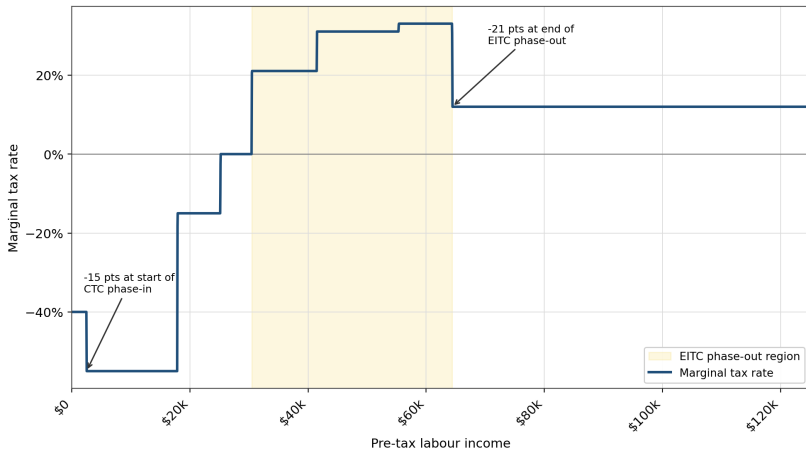
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Conclusions

- Theory and quantitative exploration of incentives to manipulate earnings risk and implications for tax design.
- When agents control the dispersion of earnings, decreasing marginal tax rates give incentives for inefficient risk manipulation.
- Marginal tax rates are often decreasing at income levels where the social transfers are phased-out.
- Removing incentives for risk manipulation leads to smaller transfers that are phased-out more slowly.

Extra figures

**Marginal tax rate (federal income tax + EITC + CTC)
Single-earner couple, 2 children, 2025**



[Return](#)

